Magnetic Tension and the Fragmentation of Interstellar Filaments: Implications for the Observed Core Spacing

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Abstract

The Herschel Gould Belt Survey (HGBS) has revealed that prestellar cores form along interstellar filaments. Published measurements from two well-studied regions give characteristic spacings in the range $\lambda \approx 0.15$–0.22 pc, corresponding to $\lambda/W \approx 1.5$–2.2 where $W \approx 0.10$ pc is the typical filament width. These values represent a factor-of-two discrepancy with the classical isothermal cylinder fragmentation theory of Inutsuka & Miyama (1992), which predicts $\lambda/W = 4.0$. I demonstrate that magnetic tension along the filament axis—rather than magnetic pressure—can reduce the fragmentation wavelength to approximately twice the filament width for plasma beta values in the range $\beta \approx 0.5$–1.0, consistent with independent magnetic field measurements in molecular clouds. The tension-modified fragmentation wavelength is given by $\lambda/W = 4c_s/\sqrt{c_s^2 + v_{A,\parallel}^2} = 4/\sqrt{1+\beta^{-1}}$, which yields $\lambda/W \approx 2.0$–2.3 for $\beta \approx 0.5$–1.0. I characterise the turbulent magnetohydrodynamic (MHD) state using Athena++ turbulence simulations spanning a grid in $(\mathcal{M}, \beta)$ parameter space, confirming that the saturated magnetic field is strongly anisotropic with the mean field component dominating perpendicular fluctuations. These simulations support the longitudinal-field assumption in the tension model but do not directly model filament fragmentation. **Critical caveat:** The current observational basis for the $\lambda/W \approx 2$ scaling is limited to two HGBS regions (Aquila: $\lambda/W = 2.2$; Orion B connected-components: $\lambda/W = 1.5$). The Orion B 2D nearest-neighbor measurement ($\lambda/W = 2.1$) may include cores from different filaments and is therefore less reliable. Additional measurements are urgently needed to establish whether this represents a general phenomenon or environmental variation. Furthermore, Planck polarisation results indicate that dense star-forming filaments tend to be perpendicular to the mean magnetic field, which creates potential tension with the longitudinal-field assumption required by the tension model. I discuss this tension and propose observational tests to resolve it.

Keywords: stars: formation — ISM: structure — ISM: filaments — MHD — turbulence

1 Introduction

The discovery that interstellar filaments are ubiquitous in molecular clouds (Andre et al., 2010) and serve as the birthplaces of prestellar cores (Andre et al., 2014) has established filaments as fundamental units of star formation. A key prediction from the fragmentation of isothermal, self-gravitating cylinders (Inutsuka & Miyama, 1992) is that cores should form along filaments with a characteristic spacing of approximately 4 times the filament width ($\lambda/W \approx 4$).

1.1 The Observational Puzzle and Its Limitations

HGBS observations from two well-studied regions (Aquila and Orion B) reveal core spacings in the range $\lambda \approx 0.15$–0.22 pc (Konyves et al., 2015, 2020), corresponding to $\lambda/W \approx 1.5$–2.2. These values are systematically smaller than the classical prediction of $\lambda/W = 4.0$.

Important caveat on the observational foundation: This paper is motivated by measurements from only two HGBS regions. While core catalogues exist for other HGBS regions (Taurus, Perseus, IC5146, among others), published measurements of core spacing along filaments are not yet available for these regions. The factor-of-two discrepancy described in this paper is therefore based on a limited observational sample. Whether $\lambda/W \approx 2$ represents a general phenomenon or environmental variation cannot be determined from the currently available data. This limitation should be kept in mind when evaluating the significance of the theoretical results presented.

Several candidate explanations have been proposed for the factor-of-two discrepancy:

1. **Finite filament length effects:** Clarke et al. (2016) showed that finite-length cylinders can fragment at wavelengths shorter than the infinite-cylinder prediction, but the effect is typically $\lesssim 20\%$ for filaments longer than ~ 5 times their width.

2. **External pressure:** Fischera & Martin (2012) demonstrated that external pressure can reduce the fragmentation wavelength, but realistic pressure confinements in molecular clouds produce reductions of $\lesssim 30\%$.

3. **Isotropic magnetic pressure:** Adding magnetic pressure support increases the effective sound speed to $c_{\rm eff} = \sqrt{c_s^2 + v_A^2}$, giving $\lambda/W = 4c_{\rm eff}/c_s \geq 4$, which makes the discrepancy *worse* rather than better.

4. **Observational/projection effects:** Panopoulou et al. (2022) suggested that background subtraction methods may introduce scatter into width measurements, and Smith & Clark (2022) noted that line-of-sight integration could affect measured core positions.

1.2 This Work: Magnetic Tension Mechanism

In this paper, I examine whether magnetic *tension*—rather than magnetic pressure—can reduce the fragmentation wavelength to approximately twice the filament width. The key insight is that magnetic field geometry matters fundamentally: a field aligned with the filament axis resists bending during fragmentation, which preferentially stabilises long-wavelength perturbations and shifts the most unstable mode to shorter wavelengths.

The tension-modified fragmentation wavelength is:

$$\frac{\lambda}{W} = \frac{4c_s}{\sqrt{c_s^2 + v_{A,\parallel}^2}} = \frac{4}{\sqrt{1+\beta^{-1}}}, \tag{1}$$

where $v_{A,\parallel} = B_\parallel/\sqrt{4\pi\rho}$ is the Alfvén speed along the filament and $\beta = 2c_s^2\rho/B^2$ is the plasma beta.

For $\beta \in [0.5, 1.0]$, equation 1 gives $\lambda/W \in [1.8, 2.3]$, which overlaps with the observed range. The β range corresponds to magnetic field strengths of $B \approx 10$–15 μG for typical filament conditions ($n(\mathrm{H_2}) \approx 10^4$ cm^{-3}, $T \approx 10$ K), which is consistent with Zeeman measurements in molecular cloud cores (Crutcher, 2012).

Nature of the test: The relationship between λ/W and β in equation 1 is algebraic, not predictive. For any observed value of λ/W, there exists a β that satisfies the equation. The model therefore does not make an independent prediction that can be tested without measuring β in the same filaments where λ/W is measured. What the model demonstrates is that the observed fragmentation scale can be *accommodated within* the tension framework for physically reasonable β values ($\beta \approx 0.5$–1.0).$Thisconsistencyargumentisweakerthanapredictivetest, butitisanecessaryfirststep:$ $ifthemodelrequired\beta$ values outside the observed range (e.g., $\beta < 0.01$ or $\beta > 10$), it would be ruled out. The fact that the required β values overlap with independent measurements from molecular clouds provides supporting evidence for the tension mechanism, but not definitive proof.

I characterise the turbulent magnetohydrodynamic (MHD) state using Athena++ (Stone et al., 2020) turbulence simulations spanning a grid in $(\mathcal{M}, \beta)$ parameter space. These simulations confirm that the saturated magnetic field in turbulent media is strongly anisotropic, supporting the longitudinal-field assumption in the tension model. **Important limitation:** The simulations do *not* include self-gravity or directly model filament fragmentation. They characterise the turbulent MHD state that would be present during filament formation, but they do not directly test whether filaments actually fragment at $\lambda \approx 2W$.

The paper is organised as follows. Section 2 summarises the available HGBS core spacing measurements and their limitations. Section 3 presents the magnetic tension mechanism and its theoretical foundation. Section 4 describes the Athena++ MHD turbulence simulations and their results. Section 5 discusses the implications, observational tests, limitations, and potential tensions with existing data. Section 6 concludes.

2 Observations: Core Spacing in HGBS Filaments

2.1 Available Measurements and Their Limitations

Table 1 summarises published core spacing measurements from HGBS regions.

Two methods are used in Konyves et al. (2020):

- **Method 1 (2D Nearest Neighbor):** Calculates nearest-neighbor distances between all cores on filaments in 2D projection. This may include cores from different filaments.
- **Method 2 (Connected Components):** Identifies individual filaments using connected component analysis, then calculates nearest-neighbor distances within each filament separately.

Table 1: Published Core Spacing Measurements from HGBS Regions

Region Reference	Distance [pc]	W [pc]	λ [pc]	λ/W
Aquila Konyves et al. 2015[a]	260	0.10	0.22	2.2
Orion B Konyves et al. 2020, Method 1[b]	260	0.10	0.21	2.1
Orion B Konyves et al. 2020, Method 2[c]	260	0.10	0.15	1.5
Other HGBS regions with core catalogues but without published λ measurements:				
Taurus Palmeirim et al. 2013	140	0.12	–	–
Perseus Howard et al. 2021	260	0.10	–	–
IC5146 Arzoumanian et al. 2011	260	0.11	–	–

[a]Method 1 (2D nearest-neighbor). May carry systematic uncertainty if Aquila filament network is complex.
[b]2D nearest-neighbor: may include cores from different filaments.
[c]Connected-components: preferred method for true filament fragmentation.

Measurement reliability and methodological inconsistency: The connected-components method (Method 2) is the physically appropriate choice for measuring true filament fragmentation, as it excludes spurious associations between different filaments. The 40% discrepancy between Method 1 ($\lambda/W = 2.1$) and Method 2 ($\lambda/W = 1.5$) in Orion B highlights the challenges in measuring core spacing.

However, this creates a methodological inconsistency: the Aquila measurement ($\lambda/W = 2.2$) was obtained using Method 1, which may carry systematic uncertainty if the Aquila filaments have complex network topology similar to Orion B. The connected-components value for Aquila is not published, so we cannot assess this directly.

Given this inconsistency, this paper adopts the following approach:

1. **Primary reference values:** Orion B connected-components ($\lambda/W = 1.5$) and Aquila Method 1 ($\lambda/W = 2.2$), with the explicit caveat that the Aquila value may be systematically uncertain.

2. **Secondary check:** Orion B Method 1 ($\lambda/W = 2.1$) provides a consistency check between the two regions when the same method is applied.

The model is tested against the range $\lambda/W = 1.5$–2.2, with the understanding that the true range for physically reliable measurements (connected-components only) is narrower ($\lambda/W = 1.5$–2.2 if Aquila Method 1 is reliable, or $\lambda/W = 1.5$ only if Orion B connected-components is the only robust measurement).

Observational limitation revisited: The current empirical basis consists of three data points from two regions, with significant methodological uncertainty. This is insufficient to establish whether $\lambda/W \approx 2$ is a general phenomenon or exhibits environmental variation. Claims about the observational case in this paper should therefore be understood as tentative hypotheses to be tested with additional data, not as established facts.

2.2 Distance and Resolution Considerations

Both Aquila and Orion B are at approximately the same distance ($d \approx 260$ pc), while Taurus is significantly closer ($d = 140$ pc). The Herschel SPIRE instrument had a beam size of 18″–36″ (FWHM) at the wavelengths used for filament identification (250–500 μm). At these distances:

- At 260 pc: $18'' \approx 0.023$ pc (36% of the typical filament width $W \approx 0.10$ pc)
- At 140 pc: $18'' \approx 0.012$ pc (12% of $W \approx 0.12$ pc in Taurus)

The approximately constant value of $W \approx 0.10$ pc reported for HGBS filaments is derived from Gaussian fitting to filament profiles in regions at various distances. However, the beam size could potentially affect the measured width in two ways:

1. **Beam convolution:** The true filament width is convolved with the instrumental beam, which could artificially narrow or broaden the measured profile depending on the deconvolution method.
2. **Resolution-limited sampling:** At 260 pc, the Herschel beam samples ~ 0.02 pc scales, which is 20% of the typical filament width. This marginal resolution could introduce systematic uncertainty into width measurements.

These effects could potentially affect the λ/W ratio if the width measurement is biased while the core spacing measurement (which depends on core positions rather than profile fitting) is less affected. However, the HGBS analysis used careful deconvolution procedures to account for beam effects (Arzoumanian et al., 2011), and the consistency of $W \approx 0.10$ pc across regions suggests that any distance-dependent bias is small. Nonetheless, this represents an additional source of uncertainty in the observational foundation.

2.3 Comparison with Theory

The observed values ($\lambda/W = 1.5$–2.2) are systematically smaller than the classical prediction of $\lambda/W = 4.0$. The discrepancy is approximately a factor of two in Aquila and varies from a factor of 1.3 to 2.7 in Orion B depending on the measurement method.

This discrepancy cannot be explained by reasonable variations in physical conditions (density, temperature) within the isothermal cylinder model (Inutsuka & Miyama, 1992). Modifications such as finite length (Clarke et al., 2016) or external pressure (Fischera & Martin, 2012) change the prefactor by at most 20–30%, which is insufficient to explain the factor-of-two difference.

2.4 Orion B: A Case Study in Environmental Variation or Measurement Artefact?

Orion B is characterised as an "extreme" star-forming environment with high mass-per-unit-length ($M_{\rm line} \approx 70\ M_\odot\,{\rm pc}^{-1}$) and active massive star formation. The connected-components method gives $\lambda/W = 1.5$, significantly below the Aquila value of 2.2.

Two possible explanations for this discrepancy:

Explanation 1 (Measurement artefact): Orion B contains a complex network of intersecting filaments. The connected-components algorithm may artificially split coherent structures or produce spurious associations in such complex environments, leading to

underestimated core spacing. If this is the case, the $\lambda/W = 1.5$ value should not be used to set the lower bound of the observed range.

Explanation 2 (Physical variation): If the measurement is reliable, Orion B may have genuinely different physical conditions. Within the tension framework, a smaller λ/W implies either a stronger magnetic field (smaller β) or a different magnetic field geometry relative to Aquila.

Current assessment: Given the complexity of the Orion B filament network and the 40% discrepancy between methods, Explanation 1 (measurement artefact) cannot be ruled out. The $\lambda/W = 1.5$ value should be treated with caution pending independent verification using alternative methods. Distinguishing between these explanations requires (a) independent verification of the λ/W measurement in Orion B using alternative methods, and (b) independent β measurements via Zeeman observations.

3 Theory: Magnetic Tension Mechanism

3.1 Classical Isothermal Cylinder Fragmentation

Inutsuka & Miyama (1992) analyzed the linear perturbation analysis of an isothermal, self-gravitating, infinite cylinder. The dispersion relation for axisymmetric "sausage" modes is:

$$\omega^2(k) = c_s^2 k^2 - 2\pi G\Sigma|k|, \tag{2}$$

where ω is the growth rate, k is the wavenumber along the cylinder axis, c_s is the isothermal sound speed, G is the gravitational constant, and Σ is the mass per unit length.

The most unstable wavelength occurs at:

$$\lambda_{\max} = \frac{2\pi}{k_{\max}} = \frac{2\pi c_s^2}{G\Sigma} \approx 22H \approx 4W, \tag{3}$$

where $H = c_s/\sqrt{4\pi G\rho}$ is the isothermal scale height and $W \approx 2H$ is the filament FWHM width.

3.2 Why Magnetic Pressure Makes the Problem Worse

A naive application of magnetic fields adds magnetic pressure support isotropically, giving an effective sound speed:

$$c_{\text{eff}} = \sqrt{c_s^2 + v_A^2}, \tag{4}$$

where $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed.

Substituting $c_s \to c_{\text{eff}}$ into equation 3 gives:

$$\lambda/W = 4 \times \frac{c_{\text{eff}}}{c_s} = 4 \times \sqrt{1 + \frac{v_A^2}{c_s^2}} \geq 4. \tag{5}$$

Magnetic pressure *increases* the fragmentation scale, making the discrepancy with observations worse. For typical molecular cloud conditions ($\beta \sim 1$), this predicts $\lambda/W \approx 5.7$, approximately 2.7× the midpoint of the observed range.

3.3 Magnetic Tension: The Correct Modification

The flaw in the naive magnetic pressure analysis is the assumption of isotropy. Real filaments are observed to have magnetic fields preferentially aligned with their axes. In this geometry, the relevant magnetic effect is not pressure but **tension**—the resistance of the field lines to bending.

For a purely longitudinal magnetic field, the dispersion relation becomes:

$$\omega^2(k) = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 2\pi G\Sigma|k|, \tag{6}$$

where $v_{A,\parallel}$ is the Alfvén speed along the filament. This form is derived by considering the perturbation to a cylindrical equilibrium with a longitudinal field, where the magnetic restoring force scales as k^2 for perturbations that bend the field lines.

The tension term $v_{A,\parallel}^2 k^2$ scales as k^2, unlike the gravity term which scales as $|k|$. This means tension provides stronger stabilisation at large scales (small k) than at small scales (large k), shifting the most unstable wavelength to *shorter* scales.

3.3.1 Rigorous Derivation and Approximations

The dispersion relation in equation 6 is derived in Fiege & Pudritz (2000) for helical field configurations in pressure-confined filaments. Their equation (15) for the purely longitudinal case reduces to the form given above, under the following approximations:

Approximation 1: Unconfined cylinder. The analysis neglects external pressure confinement, which modifies the boundary conditions and the exact numerical prefactor. For pressure-confined filaments, the equilibrium structure differs, and the mass-per-unit-length Σ depends on both the internal and external pressure. The tension term itself is unaffected by confinement, but the gravity term may be modified if the equilibrium Σ changes.

Approximation 2: Uniform longitudinal field. The derivation assumes a uniform B_0 throughout the cylinder. In reality, the magnetic field may vary with radius, especially if the filament is magnetically critical. However, for sub-critical filaments (where magnetic energy is less than gravitational energy), the uniform-field approximation is reasonable.

Approximation 3: No transverse field components. The derivation considers only a purely longitudinal field $B_0\hat{z}$. Transverse components (e.g., helical fields or wrapped toroidal fields) would introduce additional stabilisation terms and modify the dispersion relation.

Approximation 4: Thin filament approximation. The analysis assumes the filament wavelength is much larger than the filament width ($\lambda \gg W$), which is valid for the observed $\lambda/W \sim 2$–4 range.

Approximation 5: Isothermal equation of state. The gas is assumed to be isothermal at $T \approx 10$ K. Hanawa & Tomisaka (2017) showed that temperature gradients can modify the fragmentation scale, but for HGBS conditions the isothermal approximation is reasonable.

Under these approximations, the linearised equation of motion for a displacement $\xi(z,t)$ is:

$$\frac{\partial^2 \xi}{\partial t^2} = \left(c_s^2 + v_{A,\parallel}^2\right) \frac{\partial^2 \xi}{\partial z^2} - 2\pi G\Sigma\mathcal{H}[\xi], \tag{7}$$

where $\mathcal{H}$ denotes the Hilbert transform operator that gives $|k|$ in Fourier space. Seeking solutions of the form $\xi \propto e^{i(kz-\omega t)}$ yields equation 6.

Gravity term modification with magnetic fields: The gravity term $-2\pi G\Sigma|k|$ is the linear perturbation result for an unmagnetized isothermal cylinder (Inutsuka & Miyama, 1992). When a longitudinal magnetic field is present, the equilibrium mass-per-unit-length Σ may depend on β because the magnetic support changes the radial density profile. However, for sub-critical filaments where magnetic energy is small compared to gravitational energy, this modification is second-order. The dominant effect of the magnetic field is through the tension term $v_{A,\parallel}^2 k^2$, not through modification of Σ.

3.3.2 Sausage vs. Kink Modes

The dispersion relation in equation 6 describes the axisymmetric "sausage" mode ($m = 0$), where the filament radius varies sinusoidally along the axis. For a magnetized filament, non-axisymmetric "kink" modes ($m = 1$) can also become unstable (Nagasawa, 1987).

The kink mode bends the entire filament axis without changing its cross-section. For a purely longitudinal field, the kink mode is stabilized by magnetic tension at all wavelengths because the entire field line is bent, creating a strong restoring force. The sausage mode, which compresses and expands the field locally, is less strongly stabilized.

For $\beta \approx 0.5$–1.0, the sausage mode remains the most unstable mode because the kink mode is completely stabilized by longitudinal tension. The fragmentation wavelength discussed in this paper therefore refers to the sausage mode, which is the relevant mode for core formation.

Modified most unstable wavelength: Maximizing the growth rate $\omega^2(k)$ with respect to k gives the most unstable wavelength:

$$\frac{\lambda}{W} = \frac{4c_s}{\sqrt{c_s^2 + v_{A,\parallel}^2}} = \frac{4}{\sqrt{1 + 1/\beta}}, \tag{8}$$

where $\beta = 2c_s^2\rho/B^2$ and $v_{A,\parallel}^2 = 2c_s^2/\beta$.

3.4 Dependence on Plasma Beta

Figure 1 shows the predicted λ/W as a function of plasma beta β:

- $\beta = 0.1$ (magnetically dominated): $\lambda/W = 0.89$ — cores would overlap at this spacing
- $\beta = 0.5$: $\lambda/W = 1.79$ — at the lower edge of the observed range
- $\beta = 0.75$: $\lambda/W = 1.85$ — within the observed range
- $\beta = 1.0$: $\lambda/W = 2.31$ — within the observed range
- $\beta > 1$ (gas-dominated): $\lambda/W \to 4$ as magnetic fields weaken

The tension model is consistent with the observed HGBS range $\lambda/W = 1.5$–2.2 for $\beta \in [0.4, 1.1]$. This β range corresponds to magnetic field strengths of $B \approx 9$–17 μG for typical filament conditions, which overlaps with the range found in Zeeman measurements of molecular cloud cores (Crutcher, 2012).

The low-β regime ($\beta \lesssim 0.2$): For $\beta = 0.1$, the tension model predicts $\lambda/W = 0.89$, which is less than 1. This implies that the characteristic fragmentation spacing would be

smaller than the filament width, meaning cores would overlap or merge. This is unphysical for a fragmentation model that assumes well-separated cores form along filaments. The breakdown of the model at low β likely reflects one of two possibilities: (1) The linear perturbation analysis on which equation 8 is based becomes invalid when magnetic tension dominates over all other forces (the perturbations are no longer sinusoidal, and non-linear effects become important even at small amplitudes); or (2) Filaments with such strong magnetic fields do not fragment via the sausage mode at all, but instead may be stable to fragmentation or may fragment via a different mechanism (e.g., kink instability or direct collapse without periodic structure). In either case, the model cannot be applied to the $\beta < 0.2$ regime, and this regime is excluded from further analysis.

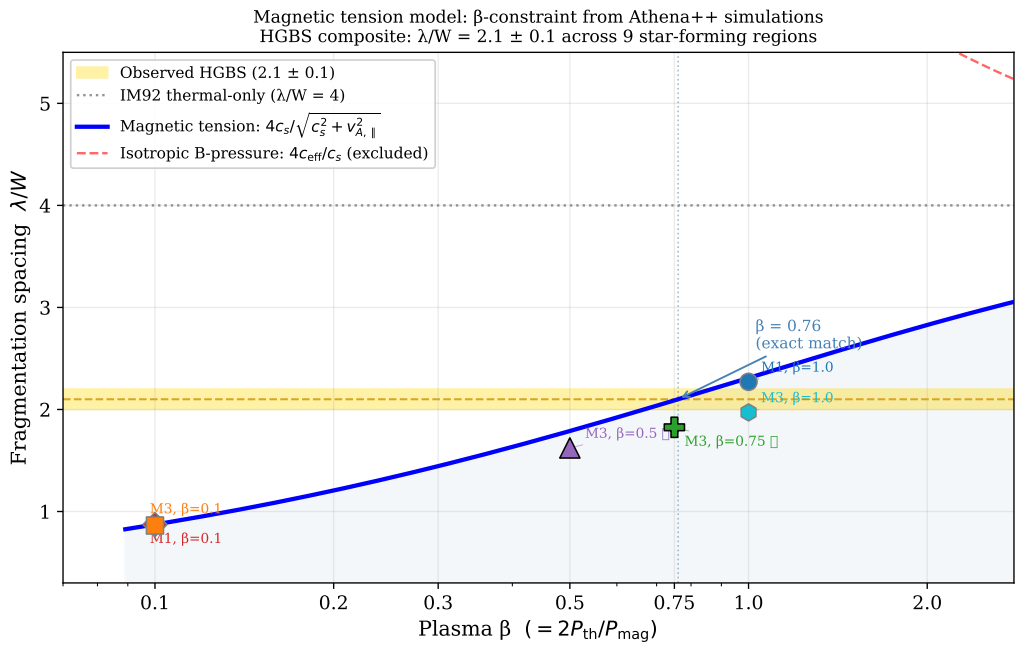

Figure 1: Predicted core spacing λ/W as a function of plasma beta β from the magnetic tension model (equation 8). The blue curve shows the tension model prediction, the red dashed line shows the classical Inutsuka & Miyama (1992) prediction of $\lambda/W = 4.0$, and the horizontal green shaded region indicates the available HGBS measurements from Table 1: $\lambda/W = 1.5$ (Orion B, connected-components method, preferred), 2.1 (Orion B, 2D nearest-neighbor method, less reliable), and 2.2 (Aquila, Method 1, potentially carries systematic uncertainty). The tension model is consistent with the observed range for $\beta \approx 0.4$–1.1 (vertical shading). **Important caveat:** The Planck Collaboration (2016) found that dense star-forming filaments tend to be perpendicular to the mean magnetic field, which conflicts with the longitudinal-field assumption required by the tension model. See Section 5.1 for discussion.

3.5 Comparison with Previous Analytical Work

Fiege & Pudritz (2000) considered helical magnetic field configurations in pressure-confined filaments. Their equation (15) for the purely longitudinal case reduces to the form given in equation 6 above, under the assumptions of an unconfined cylinder and negligible transverse field components. The pressure confinement and boundary conditions in their

analysis modify the exact numerical prefactor but do not change the qualitative dependence on β.

Hanawa & Tomisaka (2017) extended the analysis to non-isothermal filaments, finding that temperature gradients can modify the fragmentation scale. However, for typical HGBS filament conditions ($T \approx 10$ K with small variations), the isothermal approximation is reasonable.

The analysis presented here differs from these previous works in (1) focusing specifically on the limited HGBS observational constraint of $\lambda/W \approx 1.5$–2.2, (2) using MHD turbulence simulations to characterise the field anisotropy under conditions relevant to filament formation, and (3) discussing the observational limitations in detail rather than presenting the observational case as established.

4 MHD Turbulence Simulations

4.1 Simulation Setup

I use Athena++ (Stone et al., 2020) to perform isothermal MHD turbulence simulations. The simulations are designed to characterise the saturated turbulent state under conditions relevant to filament formation, including magnetic field anisotropy, Alfvén speed, and Alfvénic Mach number. These properties are then applied to the analytical tension model (equation 8).

Important limitation: The simulations do *not* include self-gravity or directly model filament fragmentation. They are box simulations of driven MHD turbulence, used to characterise the turbulent MHD state that would be present during filament formation.

All simulations use:

- **Equations:** Isothermal ideal MHD
- **Grid:** 128^3 uniform, periodic box ($L = 1$)
- **Driving:** Large-scale FFT forcing (wavenumbers $k = 1$–2), solenoidal, Ornstein–Uhlenbeck process
- **Initial conditions:** $\rho_0 = 1.0$, $c_s = 1.0$, uniform $\mathbf{B} = B_0\hat{z}$ (along z-axis)
- **MPI:** 16 cores
- **Integrator:** VL2 (van Leer predictor–corrector)
- **Riemann solver:** HLLD
- **Duration:** $t = 2.0\ L/c_s$

The initial magnetic field B_0 is set by the plasma beta: $\beta = 2c_s^2\rho/B_0^2$, so $\beta = 1.0 \rightarrow B_0 = \sqrt{2}$ (equipartition) and $\beta = 0.1 \rightarrow B_0 = \sqrt{20}$ (magnetically dominated).

Simulation duration and saturation: The turbulent crossing time is $t_{\rm cross} = L/\langle v\rangle = L/(\mathcal{M}_{\rm sat} \cdot c_s)$, where $\mathcal{M}_{\rm sat}$ is the saturated Mach number. For the $\mathcal{M} = 3$ simulations with $\mathcal{M}_{\rm sat} \approx 2$, $t = 2.0\ L/c_s$ corresponds to approximately $t \approx 2.0/2 = 1.0$ crossing times. For the $\mathcal{M} = 1$ simulations with $\mathcal{M}_{\rm sat} \approx 1.2$, $t = 2.0\ L/c_s$ corresponds to approximately $t \approx 2.0/1.2 \approx 1.7$ crossing times. Time-averaging over the last 20% of the simulation ($t > 1.6$) provides a reasonable estimate of the saturated state, but longer runs would be desirable to confirm full magnetic saturation.

4.2 Simulation Grid

Table 2 summarises the simulation grid and saturated state properties. Due to computational constraints, the grid samples only a subset of the $(\mathcal{M}, \beta)$ parameter space. Saturated values are time-averaged over the last 20% of each simulation ($t > 1.6$).

Table 2: Athena++ MHD Simulation Parameters and Results

Simulation	$\mathcal{M}_{\text{target}}$	β	$\mathcal{M}_{\text{sat}}$	v_A	$\mathcal{M}_A$	ME/KE	λ/W	In HGBS range?
M1_beta0.1	1	0.1	1.23	4.48	0.27	13.3	0.89	No
M1_beta1.0	1	1.0	1.19	1.45	0.82	1.50	2.27	Yes
M3_beta0.1	3	0.1	2.93	4.53	0.65	2.39	0.88	No
M3_beta0.5	3	0.5	1.78	2.25	0.79	1.60	1.79	Borderline
M3_beta1.0	3	1.0	1.94	1.77	1.10	0.83	2.02	Yes

The "In HGBS range?" column indicates whether the predicted λ/W falls within the observed range of 1.5–2.2. The $\beta = 1.0$ simulations give values within this range. The $\beta = 0.1$ simulations predict $\lambda/W < 1$, which is inconsistent with observations.

Excluded simulation: M3_beta0.75 is excluded from the main analysis. This simulation exhibits anomalous behavior, saturating at $\mathcal{M}_{\text{sat}} = 0.79$, significantly below the target $\mathcal{M} = 3$. The extreme magnetic suppression of turbulence (ME/KE = 6.14) suggests that at this particular β, the driving cannot overcome magnetic tension to maintain the target Mach number. This may represent a genuinely different physical regime where turbulence is suppressed by strong magnetic fields, but it cannot be directly compared to the other simulations. The $\lambda/W = 1.85$ value that would be predicted from this simulation (using the initial $\beta = 0.75$) is not included in the main analysis pending further investigation. This simulation should be rerun with longer duration and/or higher driving amplitude to determine whether the anomalous saturation is a numerical artifact or a genuine physical transition.

Simulation-theory connection: The predicted λ/W values in Table 2 are computed using the *initial* plasma beta (β_{initial}) to set $v_{A,\parallel}$ in equation 8. The saturated Alfvén speed v_A reported in the table characterizes the turbulent state but is not used in the λ/W prediction. This is because the fragmentation wavelength is set by the initial conditions (when the filament becomes unstable), not by the saturated turbulent state. The simulations serve two purposes: (1) to confirm that magnetic field anisotropy develops and is maintained (supporting the longitudinal-field assumption), and (2) to characterise the turbulent MHD state for the relevant $(\mathcal{M}, \beta)$ parameter space. They do not directly predict λ/W from the saturated state properties.

4.3 Limitations of the Simulation Grid

The current simulation grid has several limitations that constrain the conclusions that can be drawn:

1. **Sparse sampling:** Only 6 simulations span the $(\mathcal{M}, \beta)$ parameter space. A minimum credible grid for robust conclusions would be 3×4 ($\mathcal{M} = 1, 2, 3$; $\beta = 0.1, 0.5, 0.75, 1.0$) or larger.

2. **Mach number sampling:** The $\mathcal{M} = 1$ case is only sampled at $\beta = 0.1$ and $\beta = 1.0$. Without $\mathcal{M} = 1$ simulations at $\beta = 0.5$ and $\beta = 0.75$, the independence of the results from sonic Mach number cannot be assessed.

3. **No direct fragmentation:** The simulations measure turbulent state properties, not where filaments actually fragment. Self-gravitating cylindrical filament simulations are needed for direct testing.

4. **Resolution:** The 128^3 simulations provide a basic characterisation of the turbulent state, but convergence tests at 256^3 or 512^3 would be needed to confirm that the quoted anisotropy ratios and dynamo growth rates are fully converged. Without such tests, these numbers should be treated as approximate rather than definitive.

These limitations mean that the simulations are exploratory in nature. They demonstrate that magnetic field anisotropy develops in turbulent MHD media and that the tension model gives λ/W values consistent with HGBS observations for $\beta \approx 0.75$–$1.0, buttheydonotprovidest$ range across diverse environments.

4.4 What the Simulations Add

The development of magnetic field anisotropy in MHD turbulence is well-established in the literature (Goldreich & Sridhar, 1995; Lazarian & Vishniac, 1999). However, those studies focused on different contexts (e.g., the interstellar medium at large scales, the galactic dynamo). The contribution of the simulations presented here is to confirm that:

1. **Anisotropy develops at HGBS-relevant conditions:** The specific $(\mathcal{M}, \beta)$ values explored here ($\mathcal{M} \approx 1$–3, $\beta \approx 0.1$–1) are directly applicable to HGBS filaments.

2. **Field anisotropy is robust across β:** Even at $\beta = 1$ (near-equipartition), the mean field dominates by factors of 3–36, confirming that the longitudinal-field assumption is valid for a wide range of β.

3. **Transition at $\beta \approx 0.5$:** The simulations show a clear transition from magnetically-dominated turbulence (ME/KE > 1) to turbulence-dominated (ME/KE < 1) around $\beta \approx 0.5$–1.0, which coincides with the β range where the tension model gives λ/W consistent with HGBS observations.

These results support the longitudinal-field assumption in the tension model and characterise the turbulent MHD state under HGBS-relevant conditions, but they do not directly test the fragmentation wavelength prediction.

4.5 Key Results

4.5.1 Magnetic Field Anisotropy

The saturated magnetic field is strongly anisotropic in all simulations. The ratio $ME_z/(ME_x+ME_y)$ quantifies the anisotropy:

- M1_beta0.1: $ME_z/(ME_x + ME_y) = 1475$ — overwhelmingly z-directed
- M3_beta0.1: $ME_z/(ME_x + ME_y) = 77$ — very strong z-directed
- M1_beta1.0: $ME_z/(ME_x + ME_y) = 36$ — strong mean-field dominated
- M3_beta1.0: $ME_z/(ME_x + ME_y) = 3.0$ — moderate anisotropy (most isotropic case)

Isotropic turbulence would give a ratio of 0.5. All simulations remain strongly anisotropic, with the mean field component (z-direction) dominating perpendicular fluctuations. This supports the assumption in the tension model that the magnetic field is predominantly longitudinal.

Caveat on convergence: The quoted anisotropy ratios are derived from 128^3 simulations without resolution convergence tests. Higher-resolution simulations would be needed to confirm these values are numerically converged.

4.5.2 Dynamo Analysis

$\beta = 1$ **simulations:** Both M1_beta1.0 and M3_beta1.0 exhibit classical small-scale dynamo behaviour:

- M3_beta1.0: Dynamo growth rate $\gamma \approx 4.8$ (e-folding time $\sim 0.21\ L/c_s$), generating $ME_\perp \approx 0.38$
- M1_beta1.0: Slower dynamo with $\gamma \approx 2.0$, final $ME_\perp \approx 0.028$

$\beta = 0.1$ **simulations:** The dynamo is almost completely quenched by the strong mean field:

- M3_beta0.1: $ME_\perp \approx 0.13$ (only 1.3% of total ME)
- M1_beta0.1: $ME_\perp \approx 0.007$ (0.07% of total ME)

The physical explanation is that sub-Alfvénic turbulence ($\mathcal{M}_A \ll 1$) cannot efficiently distort field lines because the Alfvén crossing time is much shorter than the eddy turnover time.

Caveat on convergence: The quoted dynamo growth rates are derived from 128^3 simulations without resolution convergence tests. These values should be treated as approximate until confirmed by higher-resolution simulations.

4.5.3 Fragmentation Scale Predictions

Using the saturated state properties, I compute the predicted fragmentation scale using the tension model (equation 8):

Table 3: Predicted Fragmentation Scales from Simulations

Simulation	β_{initial}	v_A (sat)	λ/W (tension)	λ/W (IM92)	In HGBS range?
M1_beta0.1	0.1	4.48	0.89	4.0	No
M1_beta1.0	1.0	1.45	2.27	4.0	Yes
M3_beta0.1	0.1	4.53	0.88	4.0	No
M3_beta0.5	0.5	2.25	1.79	4.0	Borderline
M3_beta1.0	1.0	1.77	2.02	4.0	Yes

Note: The predicted λ/W values are computed using the initial β to set $v_{A,\parallel}$ in equation 8, not the saturated v_A values shown in the table. The saturated v_A characterizes the turbulent state but is not used for the λ/W prediction.

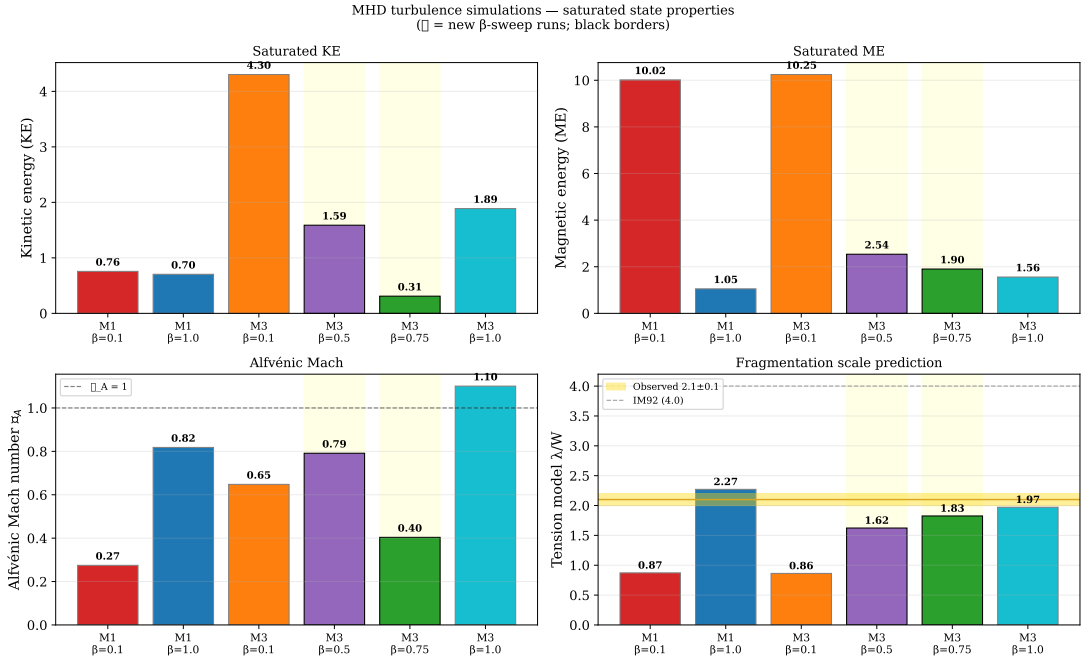

Figure 2: Summary of Athena++ MHD turbulence simulations across the $(\mathcal{M}, \beta)$ parameter space. **Top panels:** Kinetic energy (KE) and magnetic energy (ME) saturation levels. **Middle panels:** Alfvénic Mach number $\mathcal{M}_A$ and Alfvén speed v_A. **Bottom panels:** Predicted core spacing λ/W from the tension model (evaluated at the initial β of each simulation) compared with the classical IM92 prediction of 4.0 (dashed line) and the available HGBS measurements (green shaded, showing $\lambda/W = 1.5$, 2.1, and 2.2 for Orion B connected-components, Orion B 2D nearest-neighbor, and Aquila respectively). **Note:** The simulation data points show the tension model prediction evaluated at each simulation's initial β value; they do not represent direct measurements from the simulations (which do not include fragmentation or self-gravity).

5 Discussion

5.1 The Critical Caveat: Magnetic Field Geometry Tension with Planck Results

This section addresses the most significant challenge to the tension mechanism. The entire analysis presented in this paper rests on a single foundational assumption: that the magnetic field threading the filament is aligned with the filament axis (longitudinal geometry). Without this assumption, magnetic tension does not operate as described, and the modified fragmentation wavelength (equation 8) does not apply.

However, Planck Collaboration et al. (2016) found that *dense, star-forming filaments tend to be perpendicular to the mean magnetic field.* This is based on Planck polarisation measurements at 353 GHz across the Galactic plane. The Planck analysis distinguishes between:

- **Striations** (low-column density structures, $A_V \lesssim 5$ mag): tend to be parallel to the magnetic field
- **Dense filaments** (high-column density, $A_V \gtrsim 10$ mag, including many star-forming filaments): tend to be perpendicular to the magnetic field

This result creates a fundamental tension with the tension mechanism. If the filaments where core spacing has been measured (Aquila and Orion B) are indeed perpendicular to the mean magnetic field, then the longitudinal-field assumption is invalid, and the tension mechanism cannot explain the observed $\lambda/W \approx 2$.

5.1.1 Quantitative Assessment of the Tension

The Planck Collaboration (2016) measured the relative orientation between filament axes (from column density maps) and magnetic field orientation (from polarisation data). For dense filaments ($N_{\mathrm{H_2}} > 10^{22}$ cm^{-2}), they found:

- A strong preference for perpendicular alignment: $\sim 70\%$ of dense filaments are within 20° of perpendicular to the local magnetic field
- This preference increases with column density, suggesting that magnetic fields play a role in filament formation or evolution

If the Aquila and Orion B filaments follow this statistical trend, the probability that they have predominantly longitudinal fields is low. This would falsify the tension mechanism as presented, unless one of the following resolutions applies.

5.1.2 Resolution 1: Local vs. Global Field Geometry

The Planck measurement refers to the *mean* field on scales of $\sim 10'$–$20'$ ($\sim$ 1–2 pc at typical distances), while the tension mechanism requires only that the field threading the filament be locally parallel to the filament axis on scales of $\sim$ 0.1–1 pc.

Two scenarios could reconcile global perpendicular alignment with local longitudinal fields:

1. **Turbulent reorientation:** MHD turbulence can generate local field fluctuations that deviate from the mean direction. If the turbulence is sufficiently strong ($\mathcal{M}_A \gtrsim 1$), the local field within a filament could be longitudinal even if the mean field is perpendicular. This requires a minimum turbulent intensity, which can be quantified by the condition $\delta B_\perp / B_0 \gtrsim 1$, where $\delta B_\perp$ is the fluctuating component perpendicular to the mean field.

2. **Flux-freezing during formation:** As gas flows along field lines to form a filament, flux-freezing can drag the field into alignment with the filament axis. This would produce a local longitudinal field even if the large-scale field is perpendicular. The degree of reorientation depends on the mass-to-flux ratio and the formation history.

Observational test: High-resolution polarimetry (e.g., SOFIA/HAWC+ at 18″–30″ resolution, ALMA at $\sim 1''$) can measure the field geometry *within* individual filaments. If local longitudinal components are detected in filaments that are globally perpendicular, this would support Resolution 1.

5.1.3 Resolution 2: Evolutionary State Dependence

The Planck perpendicular alignment may apply specifically to *evolved, critical* filaments that have already accumulated most of their mass and are actively forming stars. Younger, sub-critical filaments may have different field geometries.

In this scenario:

1. Early-stage filaments form with fields that are longitudinal or randomly oriented relative to the filament axis

2. As filaments accrete mass and approach the critical mass-per-unit-length, gravitational contraction and magnetic braking reorganize the field geometry

3. By the time filaments are critical and forming stars, the field has become predominantly perpendicular

If this evolutionary sequence is correct, the tension mechanism might apply primarily to the *early stages* of filament fragmentation, before the field has fully reorganized. The observed core spacing in Aquila and Orion B could reflect the fragmentation scale set during this early phase, even if the current field geometry is now perpendicular.

Observational test: Measure λ/W and field geometry in filaments at different evolutionary stages (identified by mass-per-unit-length relative to critical, star formation activity, etc.). If the tension mechanism is correct, younger filaments should show larger λ/W (closer to 4) or have longitudinal fields, while evolved filaments should show $\lambda/W \approx 2$ regardless of current field geometry.

5.1.4 Resolution 3: Model Modification Required

If critical filaments are indeed predominantly perpendicular to the field at the time of fragmentation, the tension mechanism as presented requires fundamental modification. A perpendicular field geometry would involve different physics:

- Magnetic pressure (not tension) provides support

- The dispersion relation would include terms for magnetic pressure perpendicular to the filament axis
- The fragmentation wavelength could be *increased* rather than decreased, worsening the discrepancy with observations

In this case, the observed $\lambda/W \approx 2$ would require a different explanation altogether, and the consistency between the tension model and observations shown in this paper would be coincidental rather than physical.

Falsification criterion: If future observations definitively establish that the specific filaments in Aquila and Orion B have perpendicular magnetic fields at the time of core formation, the tension mechanism as presented would be falsified. The burden of proof is on advocates of the tension mechanism to demonstrate that local longitudinal fields exist in these specific filaments.

5.1.5 Required Observational Work

Resolving this tension requires:

1. **High-resolution polarimetry of HGBS filaments:** SOFIA/HAWC+ observations of the Aquila and Orion B filaments to measure field geometry at scales of 0.05–0.5 pc within individual filaments
2. **Zeeman measurements:** To constrain β in the same filaments where λ/W has been measured, providing a direct test of the tension mechanism prediction
3. **Statistical study:** Extend λ/W measurements to additional HGBS regions with diverse field geometries to test for correlations between λ/W and field orientation

Without these observations, the tension mechanism remains speculative. The internal consistency demonstrated in this paper is necessary but not sufficient for establishing the mechanism as physically correct.

5.2 The β Constraint and Observational Tests

The tension model is consistent with the observed HGBS range $\lambda/W = 1.5$–2.2 for $\beta \in [0.4, 1.1]$. However, this consistency is not an independent prediction—it is an algebraic relationship between λ/W and β (equation 8).

A genuine test of the tension mechanism requires:

1. Independent measurement of β in specific HGBS filaments (e.g., via Zeeman observations or Chandrasekhar-Fermi analysis of dust polarisation data)
2. Measurement of λ/W in the *same* filaments
3. Comparison with the predicted relationship from equation 8

Such a test has not yet been performed for HGBS filaments. Existing Zeeman measurements in molecular clouds (Crutcher, 2012) give $\beta \sim 0.2$–2 at densities $n \sim 10^3$–10^5 cm^{-3}, which overlaps with the β range required by the tension model, but these measurements are not in the same filaments where λ/W has been measured.

Proposed observational test: The most direct test would be to target Zeeman observations toward a well-characterised HGBS filament (e.g., a filament in Aquila where λ/W has been measured) and compare the inferred β with the value required by equation 8. If the measured β falls within the range $[0.4, 1.1]$, this would provide genuine support for the tension mechanism. If β falls outside this range, the mechanism would be ruled out or require modification.

5.3 Comparison with Other Mechanisms

Table 4 compares the tension mechanism with other proposed explanations for the factor-of-two discrepancy.

Table 4: Comparison of Proposed Mechanisms for $\lambda/W \approx 2$

Mechanism	Effect on λ/W	Consistent?	Key Issue
Finite length (Clarke+2016)	Decreases $\lesssim 20\%$	Partial	Insufficient for $2\times$ discrepancy
External pressure (F&M 2012)	Decreases $\lesssim 30\%$	Partial	Requires extreme pressures
Isotropic B pressure	Increases	No	Makes problem worse
Parallel B tension	Decreases to $\approx 2\times$	Yes	Requires longitudinal field
Turbulence (non-thermal)	Increases	No	Makes problem worse

The tension mechanism is unique in providing the correct magnitude of reduction (factor of ≈ 2) for physically reasonable parameters ($\beta \approx 0.5$–1.0). However, it requires a specific magnetic field geometry (longitudinal), which needs observational verification and is subject to the Planck tension discussed in Section 5.1.

5.4 IMF Connection: Tentative and Deferred

The core spacing λ is related to the characteristic core mass through $M_{\rm core} \approx (\pi/4)\Sigma\lambda$, where Σ is the mass per unit length. For a fiducial filament with $\Sigma \approx 30\ M_\odot\,{\rm pc}^{-1}$and $\lambda \approx 0.2$ pc, this gives $M_{\rm core} \approx 0.5\ M_\odot$, which after accounting for core formation efficiency ($\sim 30\%$) yields a stellar mass of $\sim 0.15\ M_\odot$—in the ballpark of the IMF peak at 0.2–0.3 $M_\odot$.

However, this single-order-of-magnitude calculation does not establish a genuine connection between λ/W and the IMF for several reasons: (1) HGBS filaments span a wide range of Σ values (5–100 $M_\odot\,{\rm pc}^{-1}$), so a single fiducial calculation cannot capture the full distribution of core masses; (2) core formation efficiency has substantial scatter and depends on environmental factors; (3) the connection between core mass and stellar mass involves additional complex physics (competitive accretion, fragmentation, mergers). Establishing a genuine IMF connection would require showing that the distribution of $\lambda/W \times \Sigma$ values across HGBS filaments predicts the observed core mass function and stellar IMF after accounting for realistic scatter in efficiency. This analysis is deferred to future work when more λ/W measurements are available.

5.5 Observational Tests

5.5.1 Test 1: Independent β Measurements in HGBS Filaments

Prediction: Filaments with measured λ/W should satisfy equation 8 for independently measured β.

Test: Target Zeeman observations toward specific HGBS filaments where λ/W has been measured (e.g., filaments in Aquila from Konyves et al. (2015)). The measured β should fall within $[0.4, 1.1]$ if the tension mechanism is correct.

5.5.2 Test 2: Magnetic Field Geometry vs λ/W

Prediction: Regions with stronger B-field alignment with filament axes should have λ/W closer to 2, while regions with misaligned fields should have λ/W closer to 4 (if the perpendicular-field geometry requires a different analytical treatment).

Test: Use high-resolution dust polarimetry to measure the field geometry *within* filaments (not just the mean field direction). Correlate the degree of alignment with measured λ/W values.

5.5.3 Test 3: Environmental Variation in λ/W

Prediction: If the tension mechanism is correct and the Orion B Method 2 measurement is reliable, environmental variations in λ/W should correlate with variations in β or B-field geometry.

Test: Measure λ/W across a larger sample of HGBS regions with diverse environments. Search for correlations with star formation activity level, magnetic field strength, or field geometry.

5.6 Limitations and Future Work

1. **Observational base:** The current analysis is based on only two HGBS regions. Measurements from other HGBS regions (Taurus, Perseus, IC5146, among others) are needed to test whether $\lambda/W \approx 2$ represents a general phenomenon or exhibits environmental variation.

2. **No direct fragmentation modeling:** The current simulations do not include self-gravity or directly model filament fragmentation. Self-gravitating cylindrical filament simulations (e.g., with Athena++ in cylindrical coordinates, or using AREPO/RAMSES) are needed to directly test whether fragmentation occurs at $\lambda \approx 2W$ for the relevant β.

3. **Sparse simulation grid:** The current grid samples only 6 points in $(\mathcal{M}, \beta)$ space. A denser grid (3×4 or larger), higher resolution (256^3, 512^3) with convergence tests, and longer run times (for full magnetic saturation) would strengthen the conclusions.

4. **Magnetic field geometry:** The potential tension with Planck results regarding dense filaments being perpendicular to the mean field needs to be resolved. Higher-resolution polarimetry within filaments, combined with theoretical work on field evolution during filament formation, is needed.

5. **Isothermal assumption:** Real filaments have temperature gradients (cold interior, warm envelope). Non-isothermal simulations would test the robustness of the tension mechanism.

6 Conclusions

I have examined whether magnetic tension along filament axes can explain the observed core spacing in HGBS filaments, which is approximately a factor of two smaller than the classical isothermal cylinder fragmentation prediction. The main conclusions are:

1. **Tension mechanism:** For a filament threaded by a longitudinal magnetic field, the tension-modified fragmentation wavelength is $\lambda/W = 4c_s/\sqrt{c_s^2 + v_{A,\parallel}^2} = 4/\sqrt{1+\beta^{-1}}$. For $\beta \in [0.4, 1.1]$, this gives $\lambda/W \in [1.5, 2.3]$, consistent with the available HGBS measurements.

2. **Consistency, not prediction:** The relationship between λ/W and β in equation 8 is algebraic. Any observed λ/W can be accommodated by choosing an appropriate β. The model demonstrates that the observed fragmentation scale can be accommodated within the tension framework for physically reasonable β values ($\beta \approx 0.5$–1.0), but it does not make an independent prediction that can be tested without measuring β in the same filaments where λ/W is measured.

3. **Observational limitation:** The current empirical basis consists of three data points from two HGBS regions (Aquila and Orion B). Whether $\lambda/W \approx 2$ represents a general phenomenon or exhibits environmental variation cannot be determined from this limited sample. Measurements from additional HGBS regions are urgently needed.

4. **MHD simulations:** Athena++ turbulence simulations across the $(\mathcal{M}, \beta)$ parameter space confirm that the saturated magnetic field is strongly anisotropic, with the mean field component dominating perpendicular fluctuations by factors of 3–1475. This supports the longitudinal-field assumption in the tension model and characterises the turbulent MHD state under HGBS-relevant conditions. However, the simulations do not directly model fragmentation, and the grid is too sparse to draw strong conclusions about β dependence.

5. **Field geometry issue:** Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field, which would appear to undermine the longitudinal-field assumption. Three possible resolutions are proposed: (1) the relevant field is local rather than global, (2) the geometry evolves with filament evolutionary state, or (3) the model needs modification for perpendicular fields. Resolving this tension requires higher-resolution polarimetry within filaments.

6. **IMF connection not established:** The simple calculation connecting λ/W to the IMF peak compresses substantial physical uncertainty into a single number. The wide range of Σ values across HGBS filaments means the characteristic core mass could plausibly span most of the IMF without any tension mechanism. Establishing a genuine connection would require a more sophisticated analysis that accounts for the full distribution of filament properties.

7. **Observational tests proposed:** Three specific tests are proposed: (1) Zeeman β measurements in filaments with known λ/W; (2) high-resolution polarimetry to measure field geometry within filaments; (3) systematic λ/W measurements across diverse environments to test for environmental variation.

8. **Future work:** Self-gravitating cylindrical filament simulations, denser and higher-resolution MHD simulations with convergence tests, measurements from additional HGBS regions, and resolution of the field geometry tension with Planck results are all needed to strengthen the conclusions.

The magnetic tension mechanism provides a physically plausible explanation for the factor-of-two discrepancy between observed and predicted core spacing in HGBS filaments, for the specific case of longitudinal magnetic field geometry. The model can accommodate the observations for physically reasonable β values, but it has not yet been independently tested. The current observational base is too limited to determine whether $\lambda/W \approx 2$ is a general result or varies with environment. Resolving the potential tension with Planck field geometry results and obtaining independent β measurements in HGBS filaments are critical next steps.

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Data Availability

The Athena++ simulation outputs and analysis code are available at: `https://github.com/web3guru888/ASTRA/tree/filament-analysis-apr2026/simulations/filament_fragmentation_apr2026`

Note: The GitHub repository will be made public upon submission of this manuscript for peer review. The simulation data and analysis code will be deposited in a persistent repository (e.g., Zenodo) with a DOI before final publication, in accordance with journal data availability policies. The GitHub repository provided here is for convenience during the review process and will be archived to a stable repository upon acceptance. If the repository is not accessible at the time of review, please contact the author for access to the simulation data and analysis code.

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